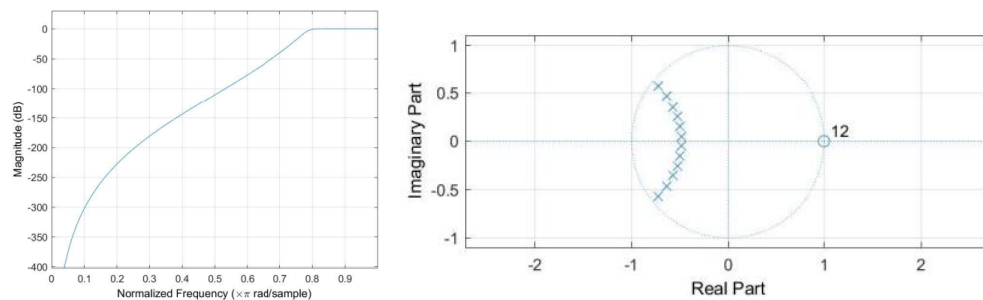


DSP Homework #15

B103012002 林凡皓

第一部分為設計一個 highpass Butterworth filter，其 normalized passband edge 在 0.8，stopband edge 在 0.7，且 stopband attenuation 為 40 dB。設計好的 filter 的架構為 direct-form II second-order sections，order 為 12。振幅響應與 pole-zero plot 如圖(一)所示，各 section 的參數設計如圖(二)所示。



圖(一)、Magnitude response and pole-zero plot of Butterworth filter

```
-----
Section #1
-----
Numerator:
 1
-2
 1
Denominator:
 1
1.45236163937257525877078023768262937665
0.850352430070693507068451945087872445593
Gain:
0.09949769792452710370778135029468103312
-----

-----
Section #2
-----
Numerator:
 1
-2
 1
Denominator:
 1
1.268938536439844666503518055833410471678
0.616665878852876825355622258939547464252
Gain:
0.086931835603258039713026050776534248143
-----

-----
Section #3
-----
Numerator:
 1
-2
 1
Denominator:
 1
1.139869718012292310049815569072961807251
0.452228320386802884467414287428255192935
Gain:
0.078089650593627629726611871774366591126
-----

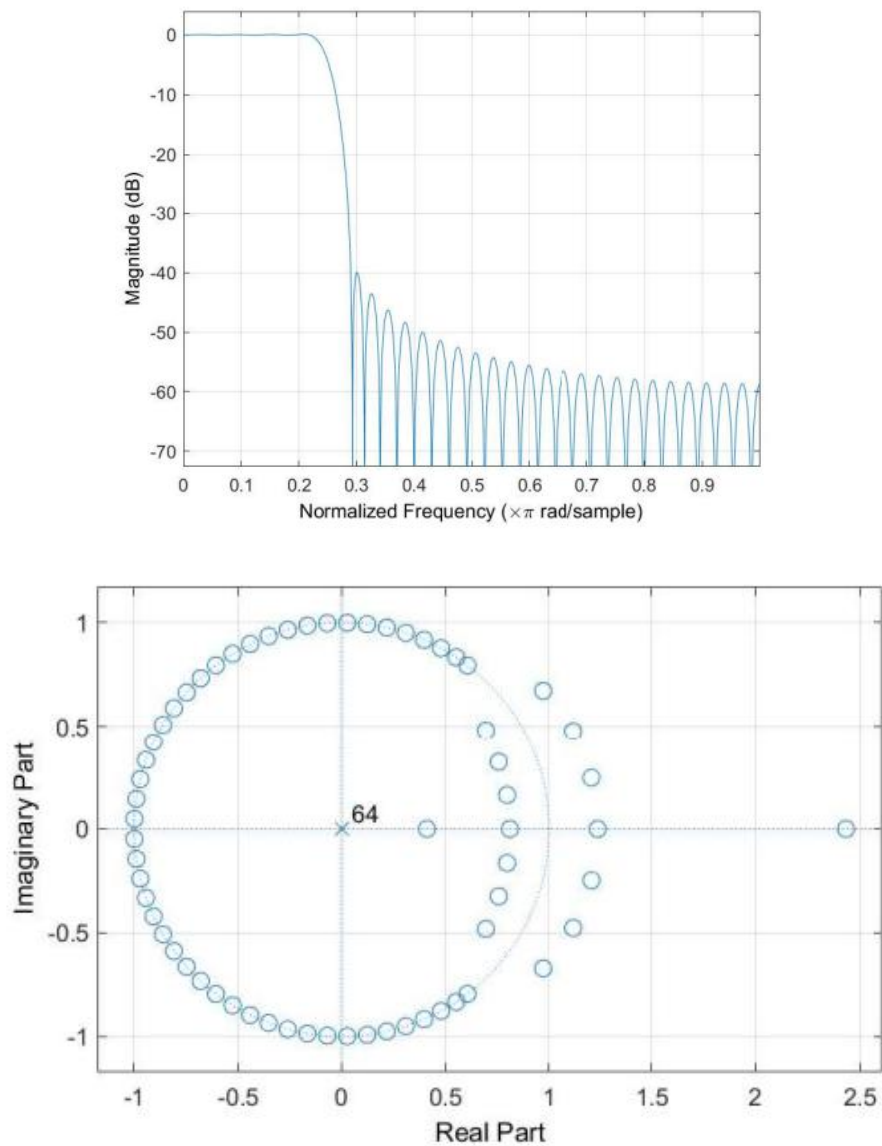
-----
Section #4
-----
Numerator:
 1
-2
 1
Denominator:
 1
1.052463531189884449190685700159519910812
0.340870208249331674288384874671464785933
Gain:
0.07210166926486180627442479362798621878
-----

-----
Section #5
-----
Numerator:
 1
-2
 1
Denominator:
 1
0.998332397245087821424647245294181630015
0.271905514752282972956720641377614811063
Gain:
0.068393279376798787883018349020858295262
-----

-----
Section #6
-----
Numerator:
 1
-2
 1
Denominator:
 1
0.972442505841958015899706424534088000655
0.238921014056074021425501996418461203575
Gain:
0.066619627053529015259236700785550056025
-----
```

圖(二)、coefficient of each section

第二部分為設計一個 lowpass Kaiser window filter，其 normalized passband edge 在 0.22，stopband edge 在 0.29，且 stopband attenuation 為 40 dB。設計好的 filter 的架構為 direct-form I，order 為 64。振幅響應與 pole-zero plot 如圖(三)所示。



圖(三)、Magnitude response and pole-zero plot of Kaiser window filter